THE LIFE PROJECT — FINAL POSTGRESQL DATABASE SCHEMA DOCUMENT (Structured for Google AI Studio)

THE LIFE PROJECT — FINAL POSTGRESQL DATABASE SCHEMA DOCUMENT

Structured for Google AI Studio

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1. DOCUMENT PURPOSE

This document defines the final PostgreSQL database schema for The Life Project.

It is aligned with:

- language: Python

- backend: FastAPI
- ORM: SQLAlchemy
- migrations: Alembic
- database: PostgreSQL
- authentication: Google Login
- phase 1 MVP scope
- AI scope: onboarding + reflection + task generation
- product direction: peace, meaning, purpose, discipline, calm, and emotional balance

This schema is designed to be:

- learnable
- relational
- clean
- expandable
- suitable for production growth

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2. DATABASE DESIGN PRINCIPLES

This database must follow these principles:

1. Clear relationships

- use proper foreign keys
- keep entity boundaries clean

1. Backend as source of truth

- frontend never owns permanent truth
- database stores real state

1. Minimal but expandable

- phase 1 keeps only essential tables

- phase 2 can add community safely

1. Separate raw inputs from interpreted outputs

- onboarding_answers stores raw user answers
- analysis_results stores system interpretation

1. Human support, not medical diagnosis

- database stores behavioral and reflective data
- not medical diagnoses

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3. PHASE 1 TABLE LIST

Phase 1 tables:

1. users
2. user_profiles
3. onboarding_sessions
4. onboarding_answers
5. analysis_results
6. tasks
7. task_logs
8. reflections
9. progress
10. tree_states
11. meditation_categories
12. meditation_recommendations
13. meditation_sessions

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4. RELATIONSHIP OVERVIEW

Core relationship map:

users → user_profiles → onboarding_sessions → tasks → reflections → progress → tree_states → meditation_recommendations → meditation_sessions

onboarding_sessions → onboarding_answers → analysis_results

tasks → task_logs

meditation_categories → meditation_recommendations → meditation_sessions

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5. MAIN ENTITY FLOW

User signs in → users row exists

User completes onboarding → onboarding_sessions row created → onboarding_answers rows created

System analyzes onboarding → analysis_results row created

System generates tasks → tasks rows created

User acts and reflects → task_logs rows created → reflections row created

System updates growth → progress updated → tree_states updated

System recommends calm support → meditation_recommendations row created → meditation_sessions row created if user starts a session

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6. TABLE: users

PURPOSE

Store the main authenticated user identity.

TABLE: users

- id UUID PRIMARY KEY
- google_sub VARCHAR(255) UNIQUE NOT NULL
- email VARCHAR(255) UNIQUE NOT NULL
- full_name VARCHAR(255) NOT NULL
- profile_image_url TEXT NULL
- auth_provider VARCHAR(50) NOT NULL DEFAULT 'google'

- is_active BOOLEAN NOT NULL DEFAULT TRUE
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()
- updated_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()
- last_login_at TIMESTAMP WITH TIME ZONE NULL

NOTES

- google_sub = stable Google user identifier
- email must be unique
- one row per user

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7. TABLE: user_profiles

PURPOSE

Store user-specific settings and baseline profile information separate from auth identity.

TABLE: user_profiles

- id UUID PRIMARY KEY
- user_id UUID NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE
- age INTEGER NULL
- occupation VARCHAR(100) NULL
- timezone VARCHAR(100) NOT NULL DEFAULT 'Asia/Kolkata'
- onboarding_completed BOOLEAN NOT NULL DEFAULT FALSE
- preferred_tone VARCHAR(50) NOT NULL DEFAULT 'calm'
- preferred_task_style VARCHAR(50) NOT NULL DEFAULT 'structured'
- primary_goal VARCHAR(255) NULL
- current_focus_area VARCHAR(100) NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

- updated_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- one-to-one with users
- stores product-level profile preferences
- separates auth from personal app settings

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8. TABLE: onboarding_sessions

PURPOSE

Represent a complete onboarding attempt or structured self-assessment session.

TABLE: onboarding_sessions

- id UUID PRIMARY KEY
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- session_status VARCHAR(50) NOT NULL DEFAULT 'completed'
- version VARCHAR(20) NOT NULL DEFAULT 'v1'
- started_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()
- completed_at TIMESTAMP WITH TIME ZONE NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- allows future re-assessment
- one user can have many onboarding sessions over time

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9. TABLE: onboarding_answers

PURPOSE

Store each individual answer given during onboarding.

TABLE: onboarding_answers

- id UUID PRIMARY KEY

- onboarding_session_id UUID NOT NULL REFERENCES onboarding_sessions(id) ON DELETE CASCADE
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- question_key VARCHAR(100) NOT NULL
- question_text TEXT NOT NULL
- category VARCHAR(100) NOT NULL
- answer_label VARCHAR(50) NOT NULL
- answer_score INTEGER NOT NULL
- question_order INTEGER NOT NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- stores raw assessment inputs
- question_key should be stable for analysis
- category examples: distraction, discipline, sleep, emotional_balance, meaning, peace, connection, digital_habits

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10. TABLE: analysis_results

PURPOSE

Store the system's interpreted result of an onboarding session.

TABLE: analysis_results

- id UUID PRIMARY KEY
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- onboarding_session_id UUID NOT NULL UNIQUE REFERENCES onboarding_sessions(id) ON DELETE CASCADE
- user_type VARCHAR(100) NOT NULL
- dominant_problem VARCHAR(100) NOT NULL
- secondary_problem VARCHAR(100) NULL

- distraction_score NUMERIC(5,2) NOT NULL DEFAULT 0
- discipline_score NUMERIC(5,2) NOT NULL DEFAULT 0
- sleep_score NUMERIC(5,2) NOT NULL DEFAULT 0
- emotional_balance_score NUMERIC(5,2) NOT NULL DEFAULT 0
- meaning_score NUMERIC(5,2) NOT NULL DEFAULT 0
- peace_score NUMERIC(5,2) NOT NULL DEFAULT 0
- connection_score NUMERIC(5,2) NOT NULL DEFAULT 0
- digital_habits_score NUMERIC(5,2) NOT NULL DEFAULT 0
- ai_summary TEXT NULL
- recommended_focus VARCHAR(100) NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- stores both rule-based scores and AI-generated summary
- one onboarding session produces one analysis result

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11. TABLE: tasks

PURPOSE

Store generated daily tasks for the user.

TABLE: tasks

- id UUID PRIMARY KEY
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- analysis_result_id UUID NULL REFERENCES analysis_results(id) ON DELETE SET NULL
- title VARCHAR(255) NOT NULL
- description TEXT NULL
- task_type VARCHAR(100) NOT NULL

- category VARCHAR(100) NOT NULL
- source_type VARCHAR(50) NOT NULL DEFAULT 'system'
- priority VARCHAR(20) NOT NULL DEFAULT 'medium'
- duration_minutes INTEGER NULL
- scheduled_date DATE NOT NULL
- status VARCHAR(50) NOT NULL DEFAULT 'pending'
- is_optional BOOLEAN NOT NULL DEFAULT FALSE
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()
- completed_at TIMESTAMP WITH TIME ZONE NULL
- skipped_at TIMESTAMP WITH TIME ZONE NULL

NOTES

- task_type examples: focus, discipline, calm, reflection, sleep, meaning
- status examples: pending, completed, skipped
- one user will have many tasks

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12. TABLE: task_logs

PURPOSE

Track task state changes over time.

TABLE: task_logs

- id UUID PRIMARY KEY
- task_id UUID NOT NULL REFERENCES tasks(id) ON DELETE CASCADE
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- action_type VARCHAR(50) NOT NULL
- note TEXT NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- action_type examples: created, viewed, completed, skipped, reopened
- useful for analytics and audit trail

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13. TABLE: reflections

PURPOSE

Store daily user reflections and related interpretation fields.

TABLE: reflections

- id UUID PRIMARY KEY
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- task_id UUID NULL REFERENCES tasks(id) ON DELETE SET NULL
- reflection_date DATE NOT NULL
- mood_label VARCHAR(50) NULL
- reflection_text TEXT NOT NULL
- main_obstacle VARCHAR(100) NULL
- emotional_tone VARCHAR(100) NULL
- ai_summary TEXT NULL
- next_step_suggestion TEXT NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()
- updated_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- one user may write many reflections
- AI should enrich, not replace, raw reflection text

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14. TABLE: progress

PURPOSE

Store the current user progress state.

TABLE: progress

- id UUID PRIMARY KEY
- user_id UUID NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE
- xp INTEGER NOT NULL DEFAULT 0
- streak INTEGER NOT NULL DEFAULT 0
- completed_task_count INTEGER NOT NULL DEFAULT 0
- skipped_task_count INTEGER NOT NULL DEFAULT 0
- reflection_count INTEGER NOT NULL DEFAULT 0
- last_active_date DATE NULL
- current_focus_area VARCHAR(100) NULL
- updated_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- one-to-one with user
- stores current summary state, not every historical event

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15. TABLE: tree_states

PURPOSE

Store the visual growth state for the tree feature.

TABLE: tree_states

- id UUID PRIMARY KEY
- user_id UUID NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE
- tree_score NUMERIC(6,2) NOT NULL DEFAULT 0
- tree_stage VARCHAR(50) NOT NULL DEFAULT 'Seed'
- growth_percentage NUMERIC(5,2) NOT NULL DEFAULT 0

- last_growth_at TIMESTAMP WITH TIME ZONE NULL
- updated_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- tree_stage examples: Seed, Plant, Young, Mature, Strong
- one-to-one with user

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16. TABLE: meditation_categories

PURPOSE

Store the meditation categories available in the app.

TABLE: meditation_categories

- id UUID PRIMARY KEY
- slug VARCHAR(50) UNIQUE NOT NULL
- title VARCHAR(100) NOT NULL
- description TEXT NULL
- default_duration_minutes INTEGER NOT NULL
- is_active BOOLEAN NOT NULL DEFAULT TRUE
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- initial categories: calm_reset, focus_reset, sleep_wind_down, emotional_reset, meaning_reflection

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17. TABLE: meditation_recommendations

PURPOSE

Store personalized meditation recommendations for a user.

TABLE: meditation_recommendations

- id UUID PRIMARY KEY

- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- meditation_category_id UUID NOT NULL REFERENCES meditation_categories(id) ON DELETE RESTRICT
- source_context VARCHAR(100) NOT NULL
- reason TEXT NULL
- recommended_duration_minutes INTEGER NOT NULL
- recommended_for_date DATE NOT NULL
- created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()

NOTES

- source_context examples: onboarding, reflection, daily_reset, sleep_support

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18. TABLE: meditation_sessions

PURPOSE

Track user meditation usage sessions.

TABLE: meditation_sessions

- id UUID PRIMARY KEY
- user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
- meditation_category_id UUID NOT NULL REFERENCES meditation_categories(id) ON DELETE RESTRICT
- meditation_recommendation_id UUID NULL REFERENCES meditation_recommendations(id) ON DELETE SET NULL
- duration_minutes INTEGER NOT NULL
- status VARCHAR(50) NOT NULL DEFAULT 'started'
- started_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT now()
- completed_at TIMESTAMP WITH TIME ZONE NULL

NOTES

- status examples: started, completed, abandoned

19. REQUIRED INDEXES

Create indexes on:

users

- email
- google_sub

onboarding_sessions

- user_id
- completed_at

onboarding_answers

- onboarding_session_id
- user_id
- category

analysis_results

- user_id
- dominant_problem
- user_type

tasks

- user_id
- scheduled_date
- status
- category

task_logs

- task_id
- user_id
- created_at

reflections

- user_id
- reflection_date

progress

- user_id

tree_states

- user_id

meditation_recommendations

- user_id
- recommended_for_date

meditation_sessions

- user_id
- started_at

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20. REQUIRED CONSTRAINTS

Important constraints:

- users.email must be unique
- users.google_sub must be unique
- user_profiles.user_id must be unique
- analysis_results.onboarding_session_id must be unique
- progress.user_id must be unique
- tree_states.user_id must be unique

Use CHECK constraints where useful:

- tasks.priority IN ('low','medium','high')
- tasks.status IN ('pending','completed','skipped')
- meditation_sessions.status IN ('started','completed','abandoned')

21. ENUM / CONTROLLED VALUE GUIDANCE

For simplicity in Phase 1, use VARCHAR + validation in FastAPI/Pydantic.

Examples of controlled values:

user_type:

- distracted_user
- undisciplined_user
- anxious_overloaded_user
- directionless_user
- low_energy_user
- balanced_inconsistent_user

dominant_problem:

- distraction
- discipline
- sleep
- emotional_balance
- meaning
- peace
- connection
- digital_habits

task_type:

- focus
- discipline
- calm
- reflection
- sleep

- meaning

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22. NORMALIZATION LOGIC

This schema keeps a practical balance between normalization and usability.

Normalized enough:

- users separate from user_profiles
- onboarding answers separate from analysis
- tasks separate from task_logs
- meditation categories separate from sessions

Not over-normalized:

- progress kept as one summary table
- tree state kept as one summary table
- reflection AI summary stored in same table for simplicity

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23. AI OUTPUT STORAGE RULE

Always store:

- raw user input
- structured rule-based scores
- AI output separately in interpretation fields

Examples:

- onboarding_answers = raw input
- analysis_results.ai_summary = AI interpretation
- reflections.reflection_text = raw reflection
- reflections.ai_summary = AI interpretation

This is critical for safety and debugging.

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24. SAFETY STORAGE GUIDANCE

Do not store:

- diagnosis labels
- medical condition claims
- harmful or extreme classifications

Allowed interpretation style:

- directionless
- overwhelmed
- distracted
- low energy
- struggling with calm

If later needed, add a separate safety_flags table.

Do not add it in Phase 1 unless you actually implement risk logic.

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25. PHASE 2 FORWARD COMPATIBILITY

This schema is ready to expand later for:

- community_circles
- circle_memberships
- circle_posts
- post_comments
- cohorts
- buddy_connections
- moderation_reports

Do not add these now.

Keep Phase 1 focused.

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26. SQLALCHEMY MODEL GUIDANCE

Each table should map to one SQLAlchemy model file.

Recommended file mapping:

- app/models/user.py
- app/models/user_profile.py
- app/models/onboarding.py
- app/models/analysis.py
- app/models/task.py
- app/models/reflection.py
- app/models/progress.py
- app/models/tree.py
- app/models/meditation.py

Keep model files clean and focused.

Do not put all models in one large file.

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27. ALEMBIC MIGRATION ORDER

Recommended migration order:

1. users
2. user_profiles
3. onboarding_sessions
4. onboarding_answers
5. analysis_results
6. tasks
7. task_logs
8. reflections
9. progress

10. tree_states
11. meditation_categories
12. meditation_recommendations
13. meditation_sessions

Reason:

Parent tables should exist before child tables.

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28. INITIAL SEED DATA

Seed these meditation categories at setup:

1. calm_reset

- title: Calm Reset
- default_duration_minutes: 5

1. focus_reset

- title: Focus Reset
- default_duration_minutes: 5

1. sleep_wind_down

- title: Sleep Wind-Down
- default_duration_minutes: 10

1. emotional_reset

- title: Emotional Reset
- default_duration_minutes: 6

1. meaning_reflection

- title: Meaning Reflection
- default_duration_minutes: 7

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29. PERFORMANCE NOTES

For Phase 1:

- PostgreSQL + indexes are enough
- do not add Redis yet
- do not add job queues yet

Correct priority:

- correctness
- clean relationships
- reliable queries

Premature performance layers will slow you down.

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30. MAIN QUERY PATTERNS THE DB MUST SUPPORT

The backend must support these common queries:

1. Get current user profile
2. Get latest onboarding session
3. Get latest analysis result
4. Get today's tasks
5. Mark task completed
6. Get latest reflection
7. Get current progress
8. Get current tree state
9. Get latest meditation recommendation
10. Get meditation categories

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31. ERD SUMMARY

ERD summary:

users1 → 1 user_profiles

1 → many onboarding_sessions

1 → many tasks

1 → many reflections

1 → 1 progress

1 → 1 tree_states

1 → many meditation_recommendations

1 → many meditation_sessions

onboarding_sessions1 → many onboarding_answers

1 → 1 analysis_results

tasks1 → many task_logs

meditation_categories1 → many meditation_recommendations

1 → many meditation_sessions

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32. FINAL JUDGMENT

This schema is:

- aligned with Python + FastAPI
- correct for PostgreSQL
- suitable for your MVP
- clean enough for learning
- expandable for future features
- safe enough for a wellbeing-focused product

This is the right database foundation for your project.

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33. NEXT STEP

The correct next document is:

FINAL BACKEND API CONTRACT DOCUMENT

Use this command next:

"Reet, generate the final backend API contract document in structured snippet format for Google AI Studio."

END OF DATABASE SCHEMA DOCUMENT